

ECON 5193: Applied Development Economics

Problem Set 2: Cluster Randomization and Spillover Effects

Spring 2026 | Due: end of Week 6

Instructions

Total Points: 100

- This problem set covers material from Weeks 4 and 5 on health interventions, with emphasis on cluster randomization and spillover effects. Submit your answers in PDF format, with your code for the empirical questions appended to the end of the PDF.
- For conceptual questions, show your reasoning. Partial credit is available.
- For empirical questions, you may use Python, R, STATA, or other statistical software. Include relevant output (tables, figures) in your submission, formatted to be easily readable.
- Data file: water_program_data.csv (available on Canvas)

Study Context

A development organization is evaluating a **School-Based Water Purification Program** designed to reduce waterborne illness and improve school attendance in rural Uganda. Diarrheal diseases are a leading cause of school absenteeism in Sub-Saharan Africa, with an estimated 1.7 billion cases of childhood diarrhea occurring globally each year.

The intervention installs water filtration systems at schools, provides chlorine tablets for water treatment, and trains teachers to deliver hygiene education. The program was implemented as a **cluster randomized controlled trial** across 60 primary schools in two districts.

The primary outcome measure is the **average monthly absence rate** at each school during the post-intervention period. The dataset includes baseline characteristics collected before randomization and geographic information to measure spillovers.

Data Description

The file `water_program_data.csv` contains the following variables:

Variable	Description
<code>school_id</code>	Unique school identifier
<code>treatment</code>	Treatment assignment (1 = water program, 0 = control)
<code>district</code>	District identifier (1 = Mukono, 2 = Jinja)

num_students	Total number of enrolled students
num_teachers	Number of teachers at school
baseline_absence_rate	Average monthly absence rate at baseline (proportion)
endline_absence_rate	Average monthly absence rate at endline (proportion) [OUTCOME]
distance_nearest_treat	Distance to nearest treatment school in kilometers
pct_female_students	Percentage of students who are female
has_latrine	School has functioning latrine (1 = yes, 0 = no)
water_source_baseline	Primary water source at baseline (1 = piped, 2 = borehole, 3 = open source)
program_cost_usd	Total program cost at school (USD) [treatment schools only]

Part I: Conceptual Questions (30 points)

Question 1. Why Cluster Randomize? (10 points)

The researchers chose to randomize at the school level rather than randomly assigning individual students within schools to treatment or control.

- (a) Explain why individual-level randomization would be problematic for this intervention. Identify at least two specific concerns. (4 points)
- (b) Define *contamination* in the context of an RCT. How would contamination bias your estimate of the treatment effect, and in which direction? (3 points)
- (c) What is the main statistical trade-off when using cluster randomization instead of individual randomization? How does this affect the power of the study? (3 points)

Question 2. Spillovers and Bias (12 points)

A key feature of this study is that treatment at one school may affect outcomes at nearby schools. For example, if children at a treated school have lower rates of waterborne illness, they may transmit fewer pathogens to children at neighboring control schools.

- (a) Define the *Stable Unit Treatment Value Assumption (SUTVA)*. How do spillover effects violate SUTVA? (4 points)
- (b) Suppose spillovers are positive (treatment at nearby schools improves outcomes at control schools). If we ignore spillovers and simply compare treatment schools to control schools, will we overestimate or underestimate the true direct effect of the program? Explain your reasoning. (4 points)
- (c) Describe the identification strategy the researchers could use to measure spillover effects in this study. What variation allows them to separately identify direct effects and spillover effects? (4 points)

Question 3. Cost-Effectiveness (8 points)

Policymakers want to know not just whether the program works, but whether it is worth the cost.

- (a) What information would you need to calculate the cost-effectiveness of this program in terms of "cost per student-day of attendance gained"? List the key components. (4 points)
- (b) Explain why including spillover effects in a cost-effectiveness calculation could substantially change policy conclusions. Use the Miguel and Kremer deworming study as an example. (4 points)

Part II: Empirical Exercises (70 points)

Use the dataset `water_program_data.csv` for the following questions. Submit your code file(s) and relevant output.

Question 4. Balance Checks at the Cluster Level (15 points)

Before analyzing treatment effects, we need to verify that randomization achieved balance across treatment and control groups.

- (a) Calculate and present the mean values of the following baseline characteristics for treatment and control schools: `num_students`, `num_teachers`, `baseline_absence_rate`, `pct_female_students`, and `has_latrine`. Present your results in a formatted table (using variable labels instead of variable names, column titles, and limiting to 3 significant digits for numerical results). (6 points)
- (b) For each variable, test whether the difference between treatment and control means is statistically significant. Report p-values. (You can simply add these to the table in part (a)). (4 points)
- (c) Based on your balance table, do you conclude that randomization was successful? If any variables show imbalance, how might this affect your analysis strategy? (3 points)
- (d) Explain why we check balance at the school level rather than the student level, even though the ultimate outcome of interest is student attendance. (2 points)

Question 5. Estimating Direct Treatment Effects (18 points)

Now estimate the direct effect of the water program on school absence rates.

- (a) Estimate the simple difference in means in `endline_absence_rate` between treatment and control schools. Report the point estimate and standard error. What is the interpretation of this coefficient? (4 points)
- (b) Now estimate the treatment effect using a regression that controls for `baseline_absence_rate` and `district` fixed effects. Present your regression output in a nicely formatted table, showing all controls. How does the point estimate compare to part (a)? Why might controlling for baseline characteristics change the estimate? (6 points)
- (c) A colleague points out that with only 60 schools, you should use robust standard errors. Explain why robust standard errors are appropriate in this context. Re-estimate the regression from part (b) with robust standard errors and discuss any changes. (5 points)
- (d) Interpret your results substantively. Is the treatment effect statistically significant? Economically meaningful? Convert the effect to student-days if helpful (assume an average school has 350 students and 20 school days per month). (3 points)

Question 6. Measuring Spillover Effects (22 points)

The key innovation in this study is the ability to measure spillover effects—how treatment at nearby schools affects outcomes at control schools.

- (a) Using **only the control schools**, examine the relationship between `distance_nearest_treat` and `endline_absence_rate`. Create a scatter plot with a fitted line. What pattern do you observe? (4 points)
- (b) Estimate a regression among control schools where the outcome is `endline_absence_rate` and the key explanatory variable is `distance_nearest_treat`. Control for `baseline_absence_rate` and `district`. Interpret the coefficient on distance. (6 points)
- (c) The coefficient from part (b) tells us how outcomes change with distance, but we want to know the magnitude of spillover effects. Calculate the predicted difference in absence rates between a control school that is 1 km from a treatment school versus one that is 10 km away. (4 points)
- (d) Compare the magnitude of the spillover effect (the benefit to a nearby control school) to the direct treatment effect you estimated in Question 5. How do they compare? What does this suggest about the importance of accounting for spillovers when evaluating this program? (5 points)
- (e) What assumption must hold for us to interpret the coefficient on `distance_nearest_treat` as a causal spillover effect? Can you think of a threat to this assumption? (3 points)

Question 7. Cost-Effectiveness Calculation (15 points)

Finally, let's calculate the cost-effectiveness of the program with and without accounting for spillovers.

- (a) Calculate the average program cost per treatment school using the `program_cost_usd` variable. Also calculate the total program cost across all treatment schools. (3 points)
- (b) Using your direct treatment effect estimate from Question 5, calculate the total student-days of attendance gained at treatment schools. Assume each treatment school has the sample average number of students and that the effect applies for a 10-month school year (200 school days). (4 points)
- (c) Now incorporate spillover effects. Using your estimates from Question 6, calculate the approximate total student-days gained at control schools due to spillovers. You may use the average distance to nearest treatment school among controls and the sample average enrollment. (4 points)
- (d) Calculate the cost per student-day of attendance gained: (i) considering only direct effects, and (ii) including both direct and spillover effects. How much does accounting for spillovers improve the cost-effectiveness ratio? Discuss what this implies for policy decisions. (4 points)

Submission Checklist

- Written answers to all questions (PDF)
- Code file(s) used for empirical questions (PDF)

- Relevant output (tables, regression results) embedded in your answers or as appendix – these should be in easily readable format, using variable labels instead of raw variable names, etc.
- Submit all files to Canvas by the deadline